João Pedro Martins Oliveira

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Education

Universidade Federal de São Paulo (UNIFESP) Expected: Jul 2027
Bachelor of Science in Computer Science *São José dos Campos, Brazil*

- **Relevant Coursework:** Software Engineering, Database Systems, Artificial Intelligence, Statistical Inference & Regression Analysis, Software Verification & Validation
- **Capstone Project / Thesis:** Building a dataset of emergency sign language in LIBRAS applied to neural networks, for a system assisting deaf individuals in contacting emergency services (SAMU, Police, Fire Department).

Professional Experience

G3P Gestão e Planejamento Financeiro Feb 2025 – Present
FullStack Software Engineer Intern *Remote*

- Engineered a desktop application integrated into Windows Explorer automating accounting package creation directly from file selection, eliminating manual web submission steps;
- Implemented automated CNPJ/CPF extraction and context-aware message suggestion from fiscal documents (NFS-e, DAS), streamlining user workflow to a single review and authorization step;
- Developed scalable asynchronous background processing pipelines using Azure Functions and RabbitMQ for decoupled accounting document ingestion;
- Architected backend services using C#/ASP.NET Core, Vue.js, SQL Server, and EF Core following Domain-Driven Design (DDD) principles, with automated testing via xUnit;

Tech Stack: C# (ASP.NET Core), Vue.js, SQL Server, EF Core, xUnit, Azure Functions, RabbitMQ, Docker, Azure DevOps, DDD

Technical Projects

Projeto wiRedPanda Aug 2023 – Present
C++/Qt Core Developer

- Co-developed the open-source digital circuit simulator wiRedPanda in C++ and Qt, adopted as the standard teaching tool for digital circuits courses at UNIFESP;
- Expanded software distribution for Linux users by creating and maintaining the official package on the Arch User Repository (AUR), streamlining installation and accessibility;
- Improved new user onboarding and project documentation by building and deploying the official web portal using Astro.js;

Tech Stack: Qt, C++, Linux, Astro.js

Goblin Jul 2024
Creator / Go Developer

- Mutation testing library written in Go as final capstone project for Software Verification & Validation at UNIFESP;
- Engineered the tool to automate synthetic syntax fault injection into the AST, uncovering blind spots and weaknesses in unit test suites;

Tech Stack: Go, AST, Mutation Testing

Technical Skills

- **Languages:** English – C1 (EF SET Certified) | Portuguese – Native | Libras – Basic
- **Programming Languages:** C++, C#, Python (Advanced) | Go, TypeScript, Java (Intermediate) | SQL, JavaScript
- **Technologies & Frameworks:** .NET/ASP.NET Core, Entity Framework Core, Azure Functions, RabbitMQ, Vue.js, Qt (C++), PyTorch, OpenCV, Astro.js
- **Databases:** SQL Server, PostgreSQL, MongoDB, SQLite
- **Engineering Practices:** DDD, Clean Architecture, Clean Code, TDD, Mutation Testing, CI/CD, Deep Learning
- **DevOps & Tools:** GNU/Linux, Docker, Git, Azure DevOps, xUnit

Certifications & Honors

- Fundamentals of Data Science – UNIFESP
- Game Developer Track – UNIFESP